

Case - Unbounded Solution

Tableau Simplex (Big-M) solution report · generated 2026-09-27 13:53

Summary flowchart branch: negative value in row 0 but no positive ratio -> the objective can grow without limit.

Model

Maximize $Z = 2X_1 + X_2$
subject to
[C1] $X_1 - X_2 \leq 10$
[C2] $2X_1 \leq 40$
 $X_1, X_2 \geq 0$

RESULT

UNBOUNDED

iterations: 2 · rule: dantzig

Findings

- Unbounded: X_3 has a negative row-0 coefficient but every entry of its column is ≤ 0 , so it can increase forever while Z keeps improving.

Contents: standard form · geometry · every tableau with entering/leaving analysis · decision diagram · summary dashboard · independent verification

1. Converting to the standard LP form

1. C1 (Type I, $\leq$): add slack $X3 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X3$ is the starting basic variable of this row.
2. C2 (Type I, $\leq$): add slack $X4 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X4$ is the starting basic variable of this row.

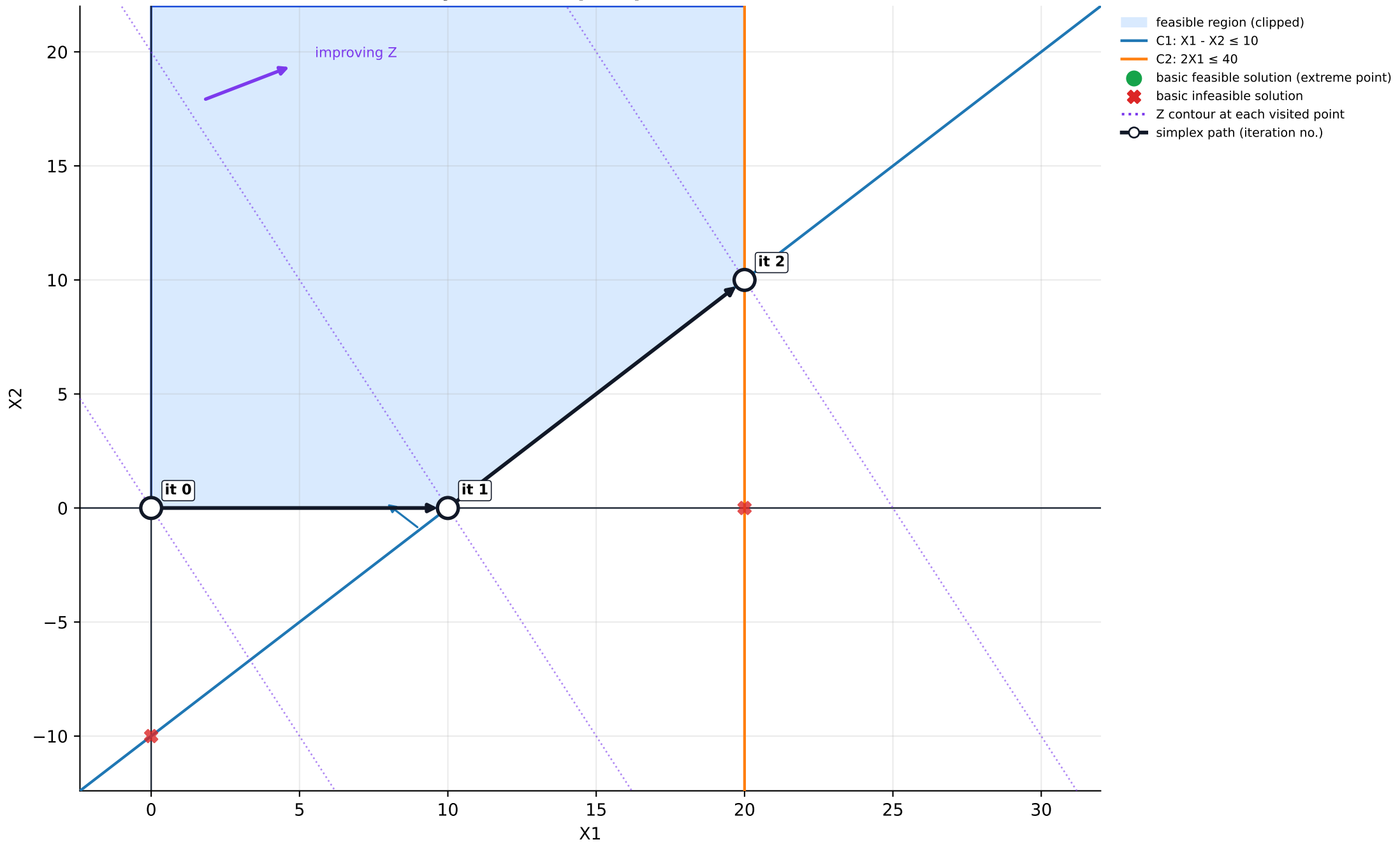
Maximize $Z = 2X1 + X2$
subject to
 $X1 - X2 + X3 = 10$
 $2X1 + X4 = 40$
 $X1, X2, X3, X4 \geq 0$

Standard form - coefficient matrix

	X1	X2	X3	X4		RHS
max Z	2	1	0	0		
C1	1	-1	1	0	=	10
C2	2	0	0	1	=	40

2. Geometric view: basic solutions, extreme points and the simplex path

Geometry of the simplex path - unbounded



3. All basic solutions (m variables basic, the rest 0). Green = feasible = extreme point; red = infeasible; ★ = optimal

Basic variables	X1	X2	Feasible?	Z	X3	X4	Optimal?
X1, X2	20	10	yes	50	0	0	★
X1, X3	20	0	no	40	-10	0	
X1, X4	10	0	yes	20	0	20	
X2, X4	0	-10	no	-10	0	40	
X3, X4	0	0	yes	0	10	40	

Initial tableau

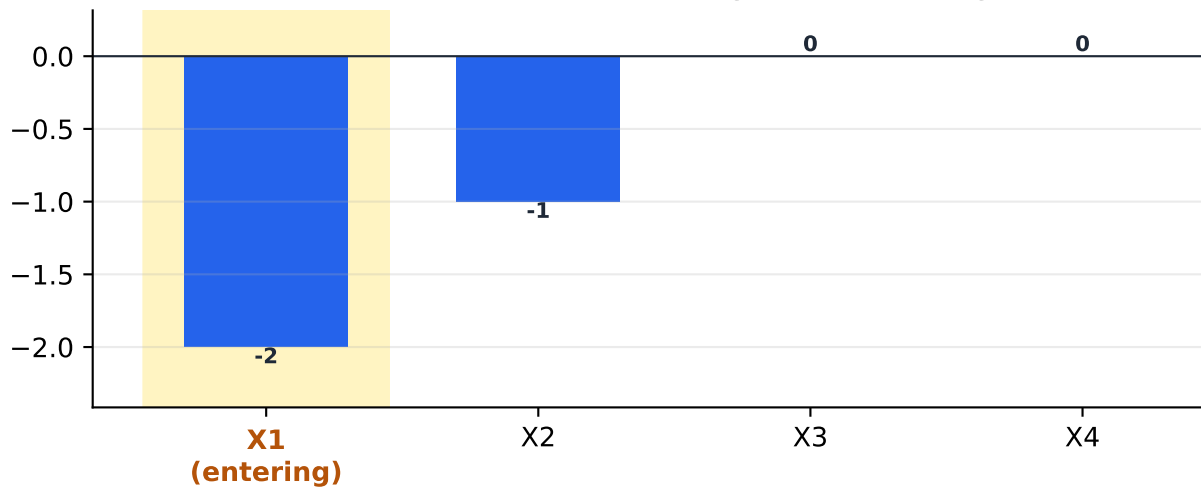
Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	-2	-1	0	0	0
X3	1	0	1	-1	1	0	10
X4	2	0	2	0	0	1	40

Initial tableau. Basic variables: $X_3 = 10$, $X_4 = 40$. Non-basic (set to 0): X_1 , X_2 . The values of the basic variables are read directly from the RHS column.

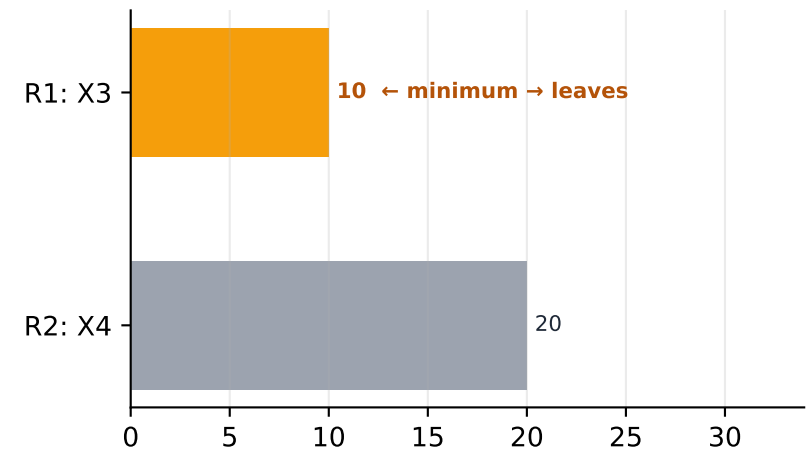
Iteration 1: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	-2	-1	0	0	0	
X3	1	0	1	-1	1	0	10	10
X4	2	0	2	0	0	1	40	20

Row 0 coefficients (most negative → entering)



Minimum ratio test (RHS ÷ X1-column)



Iteration 1 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X1 - it has the largest (in absolute value) negative coefficient in row 0 (-2).
- Minimum ratio test (RHS ÷ positive entries of the X1-column): R1: 10, R2: 20.
- Leaving basic variable: X3 (row 1, smallest ratio). Pivot number = 1.

Iteration 1: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Pivot number is already 1 - scaling not needed (normally divide the pivot row by the pivot number)

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	-2	-1	0	0	0	
X3	1	0	1	-1	1	0	10	10
X4	2	0	2	0	0	1	40	20

(b) X1 replaced X3 in the basis. Z = 20

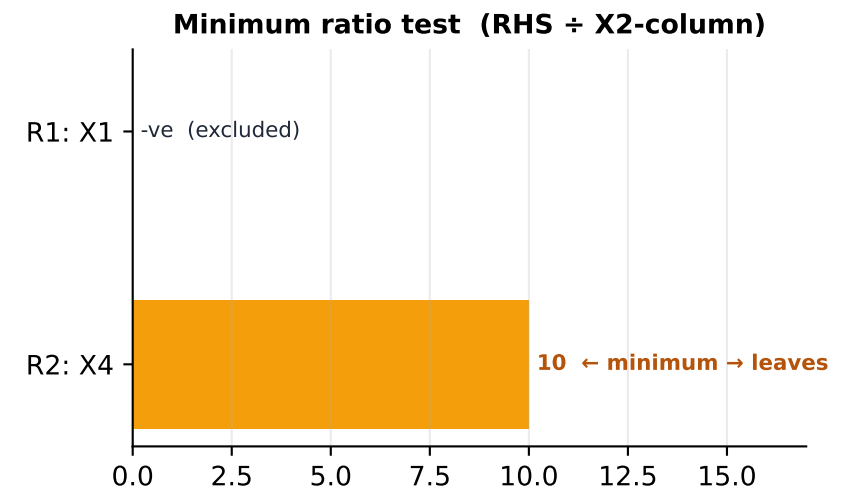
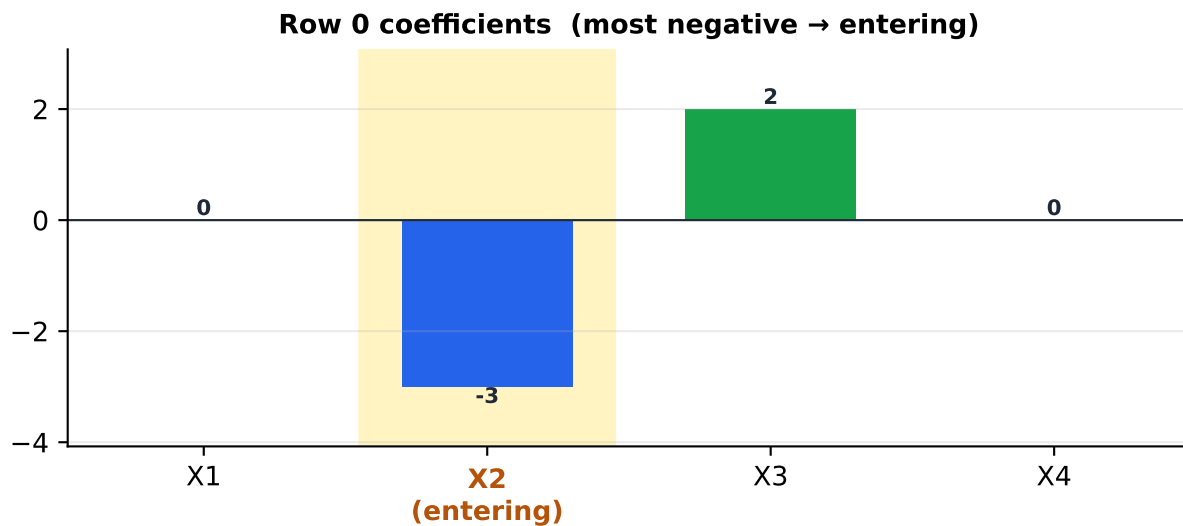
Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	0	-3	2	0	20
X1	1	0	1	-1	1	0	10
X4	2	0	0	2	-2	1	20

Row operations:

- Add $2 \times R1$ to R0
- Subtract $2 \times R1$ from R2

Iteration 2: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	0	-3	2	0	20	
X1	1	0	1	-1	1	0	10	-ve
X4	2	0	0	2	-2	1	20	10



Iteration 2 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X2 - it has the largest (in absolute value) negative coefficient in row 0 (-3).
- Minimum ratio test (RHS ÷ positive entries of the X2-column): R1: -ve, R2: 10.
- Leaving basic variable: X4 (row 2, smallest ratio). Pivot number = 2.

Iteration 2: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R2 by the pivot number 2

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	0	-3	2	0	20	
X1	1	0	1	-1	1	0	10	-ve
X4	2	0	0	1	-1	0.5	10	10

(b) X2 replaced X4 in the basis. Z = 50

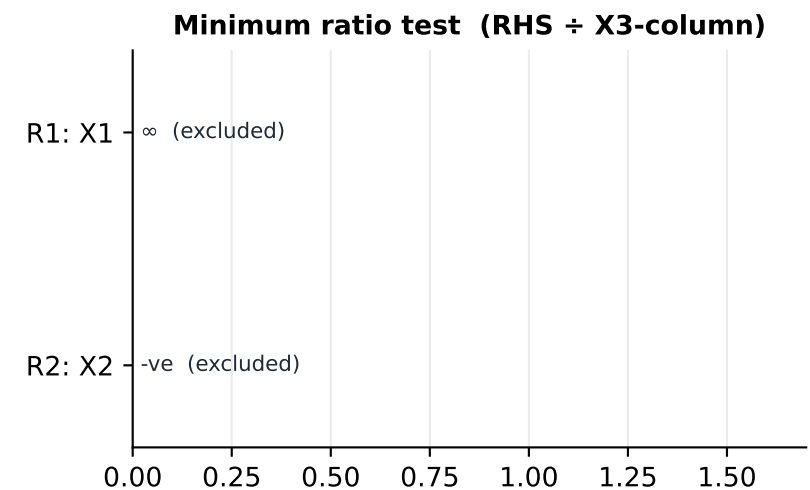
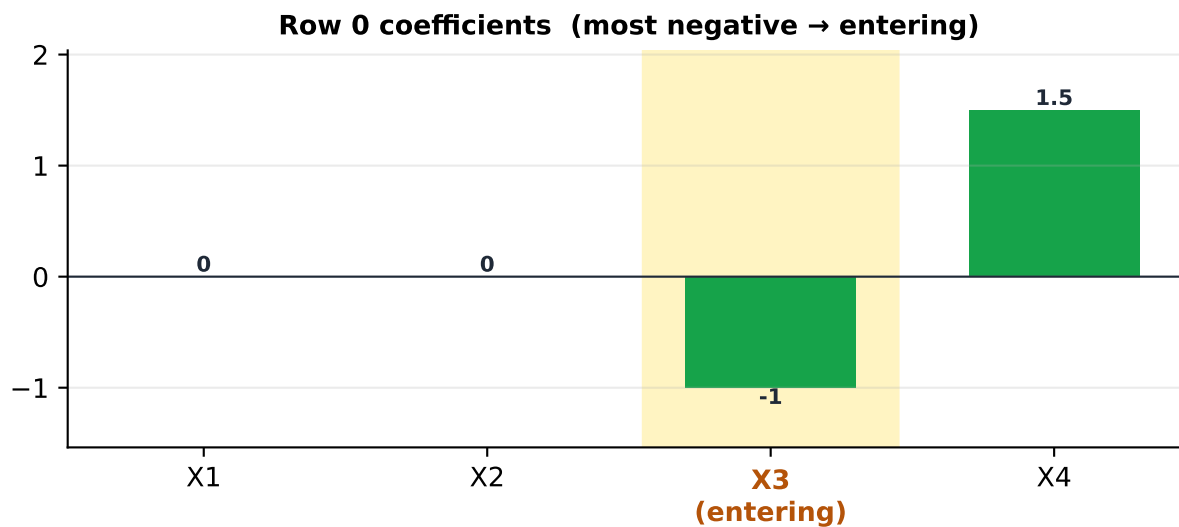
Basic	Eq.	Z	X1	X2	X3	X4	RHS
Z	0	1	0	0	-1	1.5	50
X1	1	0	1	0	0	0.5	20
X2	2	0	0	1	-1	0.5	10

Row operations:

- Add $3 \times R2$ to R0
- Add $1 \times R2$ to R1

Iteration 3: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	0	0	-1	1.5	50	
X1	1	0	1	0	0	0.5	20	∞
X2	2	0	0	1	-1	0.5	10	-ve



Iteration 3 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X3 - it has the largest (in absolute value) negative coefficient in row 0 (-1).
- No positive entry in the pivot column → no ratio → the problem is UNBOUNDED.

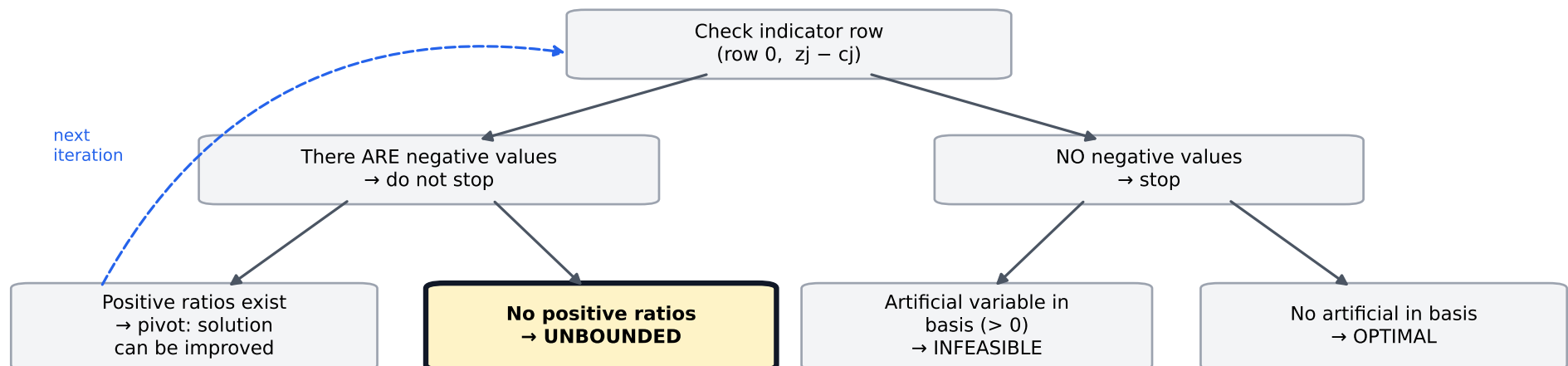
Final tableau - UNBOUNDED

Basic	Eq.	Z	X1	X2	X3	X4	RHS	Ratio
Z	0	1	0	0	-1	1.5	50	
X1	1	0	1	0	0	0.5	20	∞
X2	2	0	0	1	-1	0.5	10	-ve

Final check.

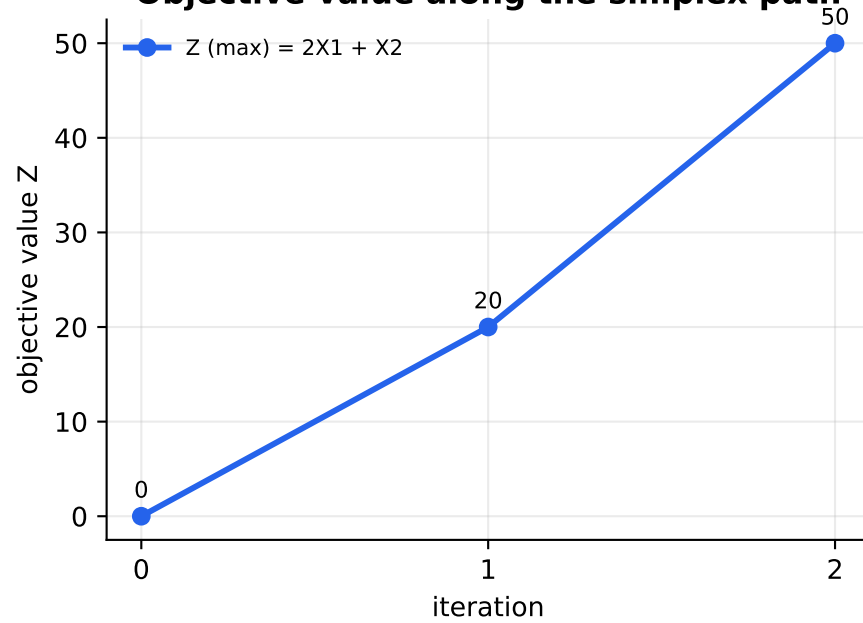
- Unbounded: X3 has a negative row-0 coefficient but every entry of its column is ≤ 0 , so it can increase forever while Z keeps improving.

Simplex decision diagram - outcome: UNBOUNDED

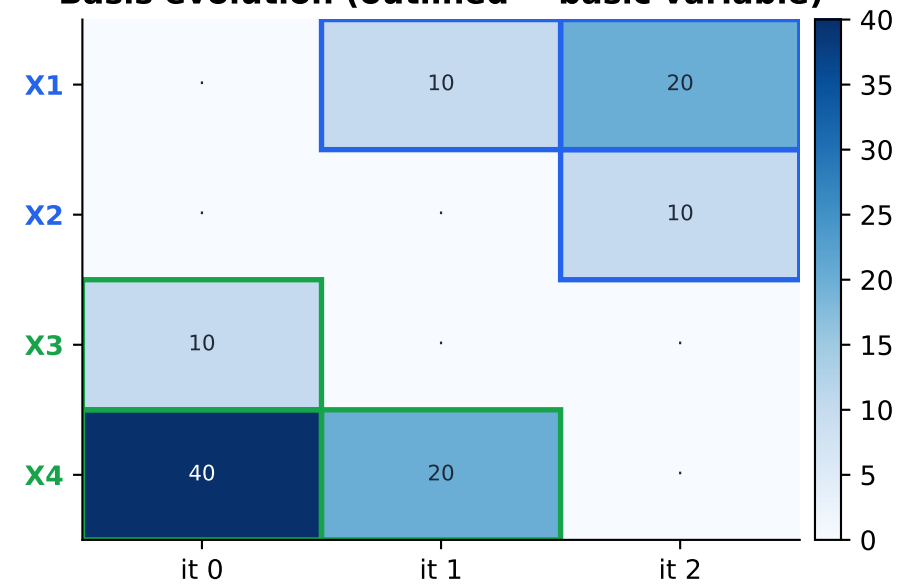


Case - Unbounded Solution - UNBOUNDED

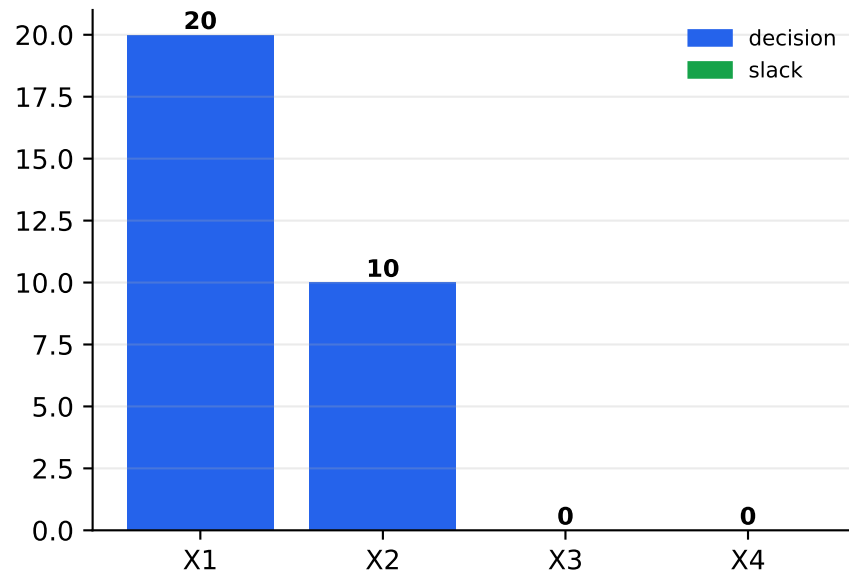
Objective value along the simplex path



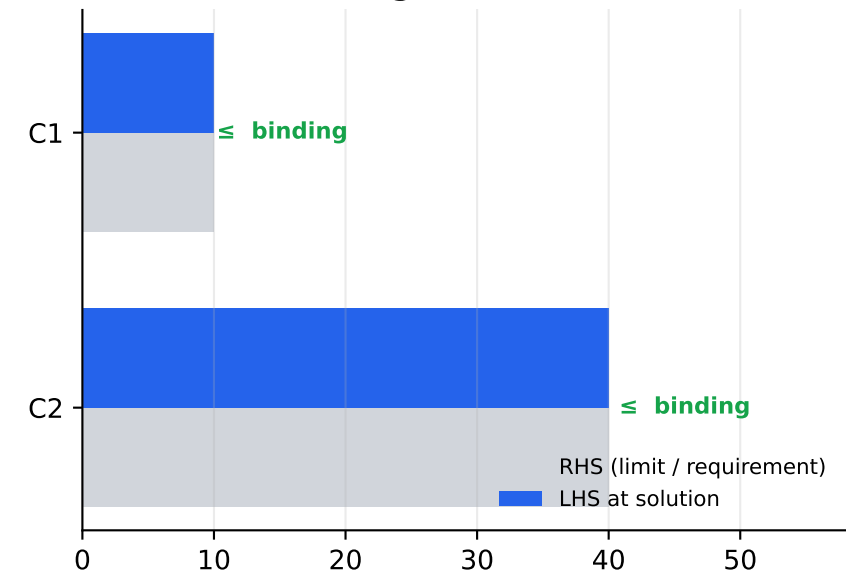
Basis evolution (outlined = basic variable)



Final values of all variables (unbounded)



Constraint usage at the final solution



Independent verification with SciPy (HiGHS)

quantity	our tableau simplex	SciPy HiGHS
status	unbounded	unbounded

✓ **Results agree**